VITALITY 09

Apomorphine

Dopamine agonist small molecule that stimulates the brain's dopamine receptors



CLASS Dopamine agonist

STUDIED FOR Parkinson's, erectile dysfunction

HUMAN EVIDENCE FDA approved (Parkinson's); ED RCTs

US STATUS FDA approved (Apokyn); not for ED

STUDIED FOR

Parkinson's off times — HUMAN

Erections — HUMAN

ED approvals ended in Europe. Uprima, Ixense and Taluvian were authorised EU-wide in 2001; all were withdrawn or expired by 2006 for commercial reasons.

HOW IT WORKS IN THE BODY

Apomorphine → Dopamine receptors → Brain signaling → Erection response

- **Binds dopamine receptors:** strongly D4; moderately D2, D3, D5 (label)
- **For ED,** studied as a drug acting in the brain, not on penile vessels
- **Sexual stimulation** was needed for Uprima to work (EMA)

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved for Parkinson's (Apokyn, 2004)
- ED: 9 placebo-controlled trials, 4 to 8 weeks
- Head-to-head trial: sildenafil did better

NOT YET ESTABLISHED

- Route of metabolism in humans

QUICK PRACTICAL INFO

HALF-LIFE About 40 minutes (range 30 to 60), under-skin injection

ROUTES STUDIED Under-skin injection or infusion; under the tongue in ED trials

PHARMACOKINETICS Peak in 10 to 60 minutes; human metabolism route not known

SOLD AS Prescription injection and infusion (Apokyn, Onapgo, generics)

MONITORED IN TRIALS Blood pressure, sleepiness, QT interval, nausea

SPORT (WADA) Not on the 2026 list

REGULATORY

FDA	Approved for Parkinson's (2004); never for ED
EMA	ED approvals (2001) withdrawn or expired by 2006
WADA	Not on the 2026 list

REFERENCES

1. FDA, Apokyn prescribing information (DailyMed), Oct 2024
2. Guillén et al., Arch Sex Behav, sublingual apomorphine for ED review, Nov 2020
3. EMA, Uprima public assessment record, May 2006

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